

Discourse Heuristics For Paradoxically Moral Self-Correction

Anonymous ACL submission

1 More Experimental Results

In this document, we would introduce more experimental results on more models such as [Phi-3.5-mini-instruct](#)(3.8b), [Llama-3-8b-instruct](#) and [Mistral-v0.3-7b-instruct](#), which provides results from both different model scope and parameter sizes. Overall, the results basically fit the findings reported in our paper.

1.1 Results of Phi-3.8b

Table 1 presents the self-correction performances of Phi-3.5-mini-instruct after finetuning on three main discourse constructions respectively. All discourses shows performance gain compared to the baseline, which prove the effectiveness of our constructions. Also, the same as the results in the paper, the discourse without statement shows best performance gain compared to others. Removing context leads to performance drop compared to All, highlighting the importance context.

Table 2 presents the in-domain and out-of domain generalization experiments for phi-3.5-mini-instruct(3.8b). The results align with that of Llama-3.2-3b presented in the paper and further demonstrate the strong capability of out-of-domain generalization in LLMs of certain sizes.

1.2 Results of larger models

This section shows experiments on larger models, Llama3-8b-instruct and Mistral-v0.3-7b. We conduct LoRA finetuning of rank 64 with learning rate of 1e-5.

Table 3 presents the experimental results for all construction settings for self-correction. Compared to the smaller models, the model with 7b/8b parameters shows better performances, especially for Llama-3-8b with four out of five perfect accuracy. Still, the proposed constructions show effectiveness for both models.

Compared with smaller LLMs, there are some

important observations: (1) Removing the Context and Action components do harm to the performance compared to the All. However, only three out of ten **All - Statement (Disability** in Llama, **Age** and **Nation** in Mistral) do harm to the performance compared to the **All**, suggesting that LLMs do not need stereotype awareness for successful self-correction. Similar to the smaller LLMs, these evidences support shallow heuristics we present in the section 4.2 of the paper. (2) Replacing Statement with Situated Statement leads to performance gain compared to baseline, while it *degrades* self-correction in the smaller models. This evidence shows more powerful capability of larger model capturing semantic within the text and provide potentials of bridging pragmatic gap via more semantically informative text.

Table 4 presents self-diagnosis performance when self-correction is enhanced. There are two key-observations: (1) The overall performance is not higher than smaller models significantly (mostly between 0.6-0.7). (2) The conflicts are mitigated a little bit in the larger models. The empirical findings ask for further fine-grained analysis for self-diagnosis, which is essential to understand how LLMs perform moral tasks.

Table 5 and Table 6 presents the in-domain and out-of-domain generalization experiments for both models. The results show strong generalization of moral self-correction in LLMs for larger models, which further extends the finding of section 5.2.

Phi3.5-mini	Age	Nation	Gender	SES	Disability
Baseline	.9176	1	.9492	.9985	.9803
All	.9574	1	.9637	.9985	.9934
All - Context	.9347	1	.9528	.9985	.9868
All - Statement	.9716	1	.9819	.9985	1

Table 1: Experimental Results on Phi-3.5-mini-instruct for **Self-correction** Across Different Discourse Constructions. **All** includes all possible components: context, statement, action. **All - *** indicates a component was removed from the setting of **All**. The optimal performance is highlighted in **bold**. Across all experiments, **All - Statement** contributes to the optimal performance in nine out of ten experiments, suggesting that LLMs do not need stereotype awareness for successful self-correction.

Phi3.5-mini	Age	Nation	Gender	SES	SexOrientation	Disability	Physical	Religion
Baseline	0.9176	1	0.9492	0.9985	0.9803	0.995	0.970	0.943
Mixed	0.9716	1	0.9691	0.9985	0.9934	0.9954	0.9824	0.9683

Table 2: Self-correction In-domain and Out-of-domain Generalization of Phi-3.5-mini-instruct By Mixing the Fine-tuning Corpus of 5 Representative Stereotypes. Phi-3.5-mini-instruct(3.8b) shows both good in-domain and out-of-domain generalization

Llama3 Mistral	Age	Nation	Gender	SES	Disability	Age	Nation	Gender	SES	Disability
Baseline	.906	.987	.955	.897	.993	.838	.837	.695	.810	.849
All	.974	1.0	1.0	.964	1.0	.878	.887	.871	.955	.882
All - Context	.926	.983	.982	.941	1.0	.869	.880	.742	.821	.862
All - Statement	.983	1.0	1.0	1.0	.938	.938	.857	.860	.988	.882
All - Action	.957	.997	1.0	.958	1.0	.847	.833	.790	.929	.875
Action	.940	.990	.969	.927	1.0	.889	.843	.735	.830	.862
All-Statement-subaction	.986	.993	1.0	.973	1.0	.898	.873	.877	.939	.875

Table 3: Experimental Results on Llama3-8b-it/Mistral-v0.3-7b-it for **Self-correction** Across Different Discourse Constructions. **All** includes all possible components: context, statement, action. **All - *** indicates a component was removed from the setting of **All**. The optimal performance is highlighted in **bold**. Across all experiments, only three out of ten **All - Statement** (**Disability** in Llama, **Age** and **Nation** in Mistral) do harm to the performance compared to the **All**, suggesting that LLMs do not need stereotype awareness for successful self-correction. **Note: We select Llama3-8b because it shares the same architecture with 1b/3b models in the paper. However, because the benchmark is popular, it have been fine-tuned with it and reach near high performance. Still, our constructions can improve it.**

Llama3 Mistral	Age	Nation	Gender	SES	Disability	Age	Nation	Gender	SES	Disability
selfdiag baseline	.665	.540	.604	.777	.697	.645	.550	.682	.635	.618
selfcorr → selfdiag	<u>.713</u>	<u>.633</u>	<u>.655</u>	<u>.882</u>	<u>.697</u>	<u>.656</u>	<u>.560</u>	.603	.600	<u>.632</u>

Table 4: Experimental Results of Llama3-8b-it/Mistral-v0.3-7b-it for Test Performance in Self-diagnosis Capability While Improving Self-correction. For all ten experiments, there are conflicts for only two of them.

Llama3 Mistral	Age	Nation	Gender	SES	Disable	Age	Nation	Gender	SES	Disable
Baseline	.906	.987	.955	.897	.993	.838	.837	.695	.810	.849
Individual	.983	1.0	1.0	1.0	.938	.938	.857	.860	.988	.882
Mixed	.992	.997	.996	.969	.993	.966	.980	.964	1.0	.967

Table 5: In-domain Generalization of Llama3-8b-it/Mistral-v0.3-7b-it By Mixing the Fine-tuning Corpus of Five Representative Stereotypes. Individual represents fine-tuning with the discourse within one stereotype in Section ?? . Mixed means that we mix the fine-tuning dataset of five stereotypes and test the fine-tuned model on each stereotype. The 3B model shows good in-domain generalization but 1B model does not, implying the generalization of heuristics does rely on model sizes.

Llama3 Mistral	SexOrientation	Physical	Religion	SexOrientation	Physical	Religion
Baseline	.947	.959	.920	.759	.849	.785
Mixed	.988	.989	.983	.951	.972	.983

Table 6: Out-of-domain Generalization of Llama3-8b-it/Mistral-v0.3-7b-it By Mixing the Fine-tuning Corpus of 5 Representative Stereotypes. 7B/8B models show good out-of-domain generalization